

	Rinsing instructions for Hydraulic systems on Loesche Mills	Rev.: 0	03-06-30 Fö/Hy																
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1. General

A competently executed assembly of all the components in the hydraulic system is a prerequisite for trouble-free future operation. All piping and fittings must be carefully checked for cleanliness, rust and burrs prior to assembly. Pipe plugs and blind flanges should only be removed shortly before final assembly. It is absolutely imperative that the equipment is not left open to the effects of the surroundings and climate. When cutting pipes, the cut must be clean, accurate and free from burrs.

Rinsing of the hydraulic circuit has an important positive effect upon the functional life of the system and its individual components (such as valves, seals and pumps).

Rinsing the hydraulic system at a high flowrate frees it from contaminative dirt and metallic deposits. This will be discussed in more detail later.

The rinsing unit comprises a tank, a pump, a filter and a measuring device (particle counter) to determine the degree of contamination of the oil.

The pump capacity depends upon the flow resistance of the piping (cross-sectional area, length and amount of fittings) as well as the viscosity and the temperature of the oil. For good results, the pump should be capable of generating 5Mpa (50 bar) at the required flowrate. The oil temperature should be at least 10°C higher than the anticipated operational temperature of the plant and be at least 50°C.

Rinsing can be carried out using the operational oil or a special rinsing oil. Using the operational oil avoids a subsequent oil change after rinsing is completed. It is necessary, however, to use a fine filter when filling the tank of the rinsing unit. The roller lubrication system does not lend itself to rinsing using its high-viscosity operational oil.

Should special rinsing oil be used, the oil must be changed after rinsing is completed.

The oil shall achieve a cleanliness of **NAS 1638 Class 8 or better**. The achieved value is to be written into a record by the company performing the rinsing.

Verification can be obtained by the following methods 

- Readout of an Online-Particle Counter
- Particle Count according to ISO 11500
- Gravimetric Examination according to ISO 4405
- Qualitative microscopic examination

Alternatively, the sample may be examined by a specialist company, such as Messers „WearCheck“. This procedure is, however, time-consuming as it involves transport of the oil to their laboratories. This may take some days.

A neutral form for a rinsing record can be found in the appendix.

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Competent cleansing of the hydraulic system represents an economic advantage for the owner of the plant.

2. Tools and equipment for the rinsing operation

2.1 Rinsing unit

The rinsing unit consists of:

- An oil tank, the volume of which is at least 5 times greater than the capacity of the hydraulic system and which is equipped with a heater, water separator and breather.
- A pump (possibly multi-staged to cope with different flowrates), with a capacity based on the following table:

Pipesize [mm]	Oil viscosity [mm ² /s]	Flowrate [l/min]	Pump power [kW]
38 x 4	68	220	24
	46	150	18
	32	110	12
30 x 3 resp. 28 x 2	68	180	22
	46	120	15
	32	90	12

- A fine filter as a duplex, switchable filter.
- An (online) particle counter in the return line – or an alternative method of determining the degree of contamination.
- Universal connections (Ermeto-fittings, SAE-Flanges, DIN-Flanges and seals), hoses, plugs etc.
- A throttle to regulate the pressure or the flowrate.
- Temperature and pressure gauges.
- Electrical connections with suitable switchgear.

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2.2 Connections

The spring systems on modern LOESCHE Mills are equipped with single or tandem hydraulic cylinders, bladder and piston-type accumulators and a ring resp. connecting pipeline executed in Ø38x4 pipe.

The roller lubrication system is executed in Ø28x2 pipe. All pipework is equipped with Ermeto fittings as standard. Occasionally welded or special fittings are used, commensurate with customer requirements. The connections between the cylinders and their accumulators are made with hoses equipped with 2" SAE-flanges or Ermeto 38-PS-fittings.

The sheer multiplicity of connection types and layouts mean that it exceeds the scope of this document to provide an overall specification for the connection components necessary for the rinsing operation.

Minimum requirements are:

- Elbow fitting W38SA3C
- Straight fitting G38SA3C
- Straight fitting G28LA3C
- Blind plug with nut VKAM38A3C (to close off open fittings)
- Closure fitting ROV38ZSA3C (to close off open pipes)
- Hose HD-3TE-DN32-DKOS-DKOS-2000 (for interconnection of Ermeto fittings)
- Hose HD-3TE-DN40-SFL3000/SFL3000-with full flange-2000-2" including
- O-Rings (for interconnection of SAE fittings)
- Hose HD-3TE-DN25-DKOL-DKOL-1000 (for interconnection of Ermeto fittings)

The required quantities will be determined according to the number of cylinders, their individual connections and the possible positions where the hydraulic circuit can be opened. This can typically only be finally decided at the site.

3. Preparation for the rinsing operation

The hydraulic system of a loesche mill consists primarily of the hydraulic cabinet(s), type HSLM and HSMS, the ring and connection line and the cylinders with their associated bladder and piston accumulators.

The drawings of the the spring assembly, ring line, connection line and lubrication lines show the location and layout of the major components. With the help of these documents a point can be chosen where the hydraulic piping can be opened to create a rinsing circuit. Typical points are to be found beneath the hydraulic cabinets and in the ring line, where 2/2 way ball valves are installed. Provisional connections can be made behind the pedestals with the aid of Ermeto fittings and medium pressure hoses.

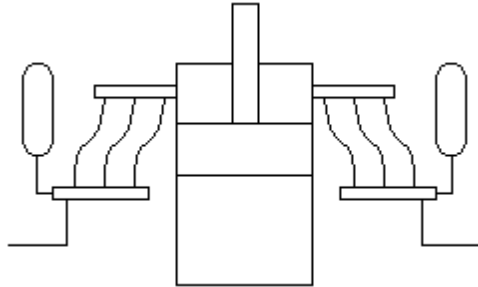
The cylinders, piston and bladder accumulators are to be excluded from the rinsing procedure. The accumulators are to be filled to 5 MPa (50 bar) to avoid the ingress of oil.

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Elbows, distributors or other components which are difficult to integrate into the rinsing circuit because of their form or position, are best cleaned individually with a free oil return to tank. The pressure loss in the piping can be taken as 0,15 MPa (1,5 bar) per 10m of pipe run. Pressure losses will also be induced by changes in the piping cross-section, bends and height differentials. These must also be considered when determining the connection of the rinsing circuit. The flowrate in the rinsing circuit is to be chosen such that the velocity in the pipes lies at around 8-10 m/s in order to achieve the best possible results. Each cabinet system is to be separately treated. A scheme for this can be found in the appendix. 4. The rinsing operation One begins with the circuit associated with the rod side of the hydraulic cylinders (see sketch in the appendix). After completion of the circuit it can be slowly filled with oil. The available pump can be used for this. Once the system has been filled, the pump flowrate can be increased. The rinsing operation is to be carried out until the desired degree of cleanliness has been achieved. It is possible that the filter must be changed during this time; it will depend upon how contaminated the original oil filling was. The degree of cleanliness can be checked with an online particle counter or another appropriate method; external analysis by an independent authority causes time losses. Thereafter the next circuit can be created (for the piston side) and the whole procedure repeated. Finally the circuits for the roller lubrication lines can be treated. Once all the piping has been successfully rinsed the working circuits can be reconnected as shown in the drawings of the spring assembly, ring and connection lines as well as the lubrication lines. Cleanliness is imperative here. The entire system can be filled with the operational oil. The nitrogen pressure in the accumulators can be adjusted to the operational values. The system can be run up to pressure and the fittings and flanges checked for leakages. The successful execution of the rinsing operation is to be documented in a record.			

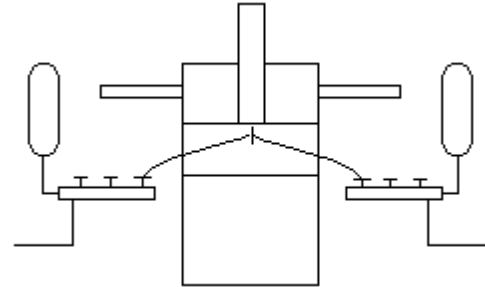
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Parallel to this, an oil sample is to be taken from the system in the neighbourhood of the cylinders and sent to an independent institute for separate chemical/mechanical analysis. This is to be compared to the analysis of the original oil. When the analyses concur and the required degree of cleanliness to NAS 8 has been achieved, the hydraulics of the machine are declared to be operational . In the event of a future oil change, care must be taken that the oil is passed through a fine filter before it enters the system. Unfiltered oil may not be taken directly from the barrel. 5. Appendix The sketches represent the arrangement principle (see next sheet).			

Quality goals

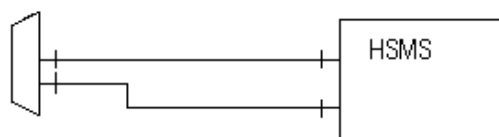
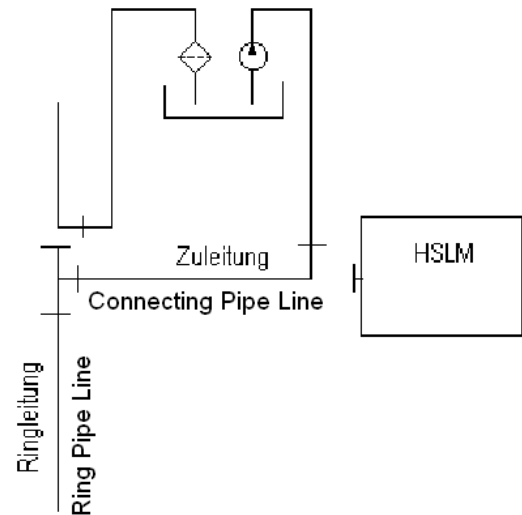
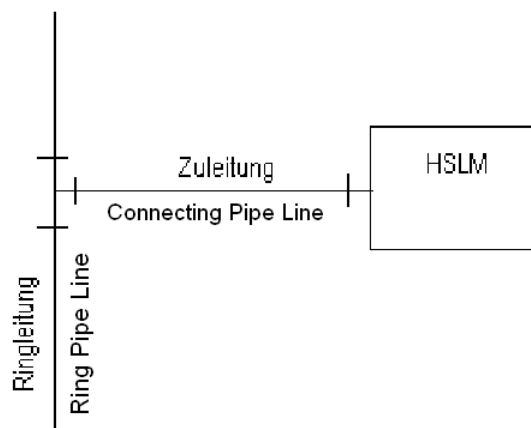
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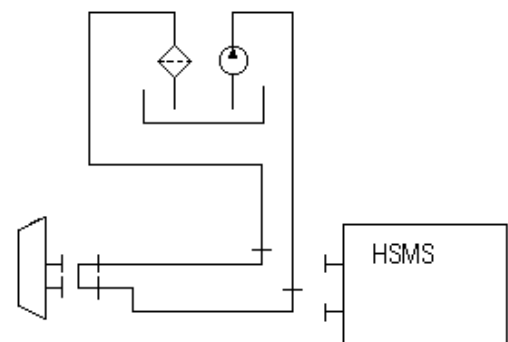
Standard connection between
HSLM and hydraulic cylinder



Connection when rinsing



Standardconnection between
HSMS und grinding roller



Connections when rinsing

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Protokoll über das Spülen des Rohrleitungssystems

Record on the flushing and cleaning of the pipe systems
Compte rendu sur le rinçage du système de tuyauteries

Lieferant der Anlage	Contractor of the unit Fournisseur de l' installation		
Betreiber	Operator Utilisateur		
Projekt	Project Projet		
Auftragsnummer	Order number Commande numero		
Die Spülarbeiten wurden fertiggestellt am The flushing work was completed on Les travaux de rinçage ont été achevés le			
Anlageteil/Spülkreislauf Part of the unit / flushing circuit Partie de L' installation / cycle de rinçage			
Medium	Fluid	Fluide	
Medium Temperatur	Medium temperature	temperature du fluide	
Aussentemperatur	ambient temperature	Temperature environnement	
geforderte Reinheitsklasse cleanliness class required classe de pureté exigée	ISO	NAS	
erreichte Reinheitsklasse cleanliness class reached classe de pureté'	ISO	NAS	
Nachweis durch Evidence by Demonstration par pièce justificative	Hydac Certifikat Online	FCU 2010 Nr. printed	yes no
Bemerkungen	Comments Remarques		
Hydac Signature contractor Signature livreur	Kunde Customer Client	Betreiber Operator Utilisateur	